

to a metaverse that becomes a shared space for retail experiences. Similarly, industrial metaverse is evolving into a virtual realm where the digital world of factories integrates into a seamless network of supply-chain operations to create a high-touch product and service experience,” explains Umang Garg, managing director, Global Practice Leader, Nagarro.

The metaverse, which according to a recent McKinsey report has the potential to generate up to \$5 trillion in value by 2030, is going to be a big force in the digital transformation of industries.

Decrypting the industrial metaverse

Emerging technologies are dynamic in nature—and it is difficult to nail them down with a globally-accepted definition. So, let us try to understand the industrial metaverse by means of its evolution rather than its definition.

Let us start with the digital twin—a technology that, according to a GE blog, has been around since 2002. It started as a product lifecycle management (PLM) tool for complex systems such as spacecrafts. In the early years, it was just a digital model of a physical system, somewhat like a prototype or blueprint, using which you could run tests and validate a design upfront before building it. Over time, the digital models were paired with specific instances or physical objects by adding data such as its dimensions, location, working conditions, etc.

Enter the Internet of Things (IoT). Sensors and actuators were added to the physical object and the data collected by these was fed to the digital twin in real time, making the digital twin a virtual mirroring of the current status of the physical object. With it, you could actually get real-time information about how a physical system is working, and even control it through the actuators. As a next step, environmental data was also fed to the digital twin, which helped

How can the metaverse help industries?

- Capture valuable data and insights from your assets and processes
- Apply artificial intelligence and machine learning to understand and optimise products and processes
- Detect and mitigate problems early, before they cause any serious impact
- Simulate scenarios like part failure, natural disasters and supply delays to understand causes and impact—and proactively prevent them
- Schedule predictive maintenance
- Collaborate with people across the world to develop and test new ideas
- Correct errors at the design stage itself
- Conduct what-if analyses using exact, working copies of products and processes, and real-time data
- Trial changes in the virtual space before implementing in real-life—compute power on the cloud enables you to try changes at jet speed, enabling multiple iterations to fine-tune the process
- Train employees in immersive virtual facilities that work exactly like the original—this is as good as hands-on training, but with less risk involved
- Enable workers to troubleshoot even complex problems by augmenting their view with required information, and also enabling an expert to connect and guide them remotely
- Make processes more sustainable:
 - Figure out the recycling puzzle
 - Develop products sustainably with minimal real-life iterations, leading to lower use of resources and lesser waste
 - Optimise processes to be power-efficient and sustainable
- Use the metaverse as a new marketing channel for customers to view or experience products remotely
- Enable business transactions through upcoming decentralised finance (DeFi) channels and digital currencies

understand how the system reacted to changes in the environment.

Fast-forward to the present. By adding intelligence to the digital twins, by way of analytics, machine learning (ML) and other algorithmic models, it has become possible to predict and prevent errors, and tune the system for maximum efficiency. With plug-and-play sensors and actuators, high-speed communications, industry-specific cloud based analytics, and intelligence tools, it is quite easy and cost-effective to implement a digital twin today.

However, the real story begins only now. These standalone digital twins must be networked to gain real, strategic advantage! Networks of digital twins help an organisation understand the interplay of various systems—how one affects the other, patterns, trends, and so on.

Now, put this network of digital twins on a private metaverse and let your employees access it through their avatars. They can, virtually, be in the scene, watch a machine

work, and even feel it heating up due to environmental factors. Derick Jose, co-founder of Flutura Decision Sciences and Analytics, explains in an interview to YourStory: “For example, we are working with energy companies that have a complex array of equipment whose health we are predicting in real time. A reliability engineer can enter the industrial metaverse, look at the subsystems of the process in the metaverse environment, and decide what is the next best action to undertake.”

This ability leads to numerous applications and benefits, such as training new users without risk, identifying and preventing errors before they happen, understanding the impact of design or process changes before implementing them, and more.

Next, bring some of your key stakeholders like vendors, collaborators, and customers also into the metaverse. What you get is a complete virtual enterprise spanning the entire value-chain. You no longer need to visualise